



**FACULTY OF COMPUTER SCIENCE AND MATHEMATICS
BACHELOR OF COMPUTER SCIENCE (SOFTWARE ENGINEERING) WITH
HONOURS**

**CSE3433 SOFTWARE ARCHITECTURE
SEMESTER II SESSION 25/26**

PROJECT TITLE:

**TALENTFLOW – EVENT-DRIVEN JOB RECRUITMENT & APPLICATION
PROCESSING SYSTEM**

PREPARED FOR:

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Selected Architecture Style

The project system TalentFlow implements Event-Driven Architecture (EDA) through its publish-subscribe system. The system does not wait for incoming requests because it responds to events which occur in the system.

Main Components:

- Event Producers
 - These users include candidates and recruiters. An event will be created when they perform any action such as submitting an application.

- Event Consumers
 - These services listen for events which they use to trigger automatic actions such as data updates and notification delivery.

- Event Bus
 - The system functions as a central hub which collects all incoming events and directs them to their appropriate destination services. The system uses this feature to maintain its operational stability during peak user traffic times.

Why we choose this architecture:

- Less dependency between components
 - Each part works separately, so if one part has an issue, others can still function.
- Better performance for large data
 - It can handle many job applications at once without slowing down the system.

Problem Description

Nowadays, many recruitment systems are still manual, and this causes several problems:

- **Hiring process is slow**
HR needs to check resumes one by one and contact candidates manually.
- **No immediate response**
When candidates apply, nothing happens instantly, and they need to wait for HR to process it.
- **Communication is not efficient**
Candidates may wait too long for updates and might lose interest.
- **High chance of mistakes**
Manual work can lead to errors like missing candidate data.
- **Hard to manage many applications**
When there are too many applicants, it becomes difficult for HR to handle everything.

System Objectives

The main goals of TalentFlow are:

1. To automate the recruitment process by triggering actions when events happen (like job posting or application submission).
2. To reduce the workload of HR by automating tasks such as filtering resumes and managing applications.
3. To improve communication by sending real-time updates to candidates.
4. To reduce human errors by organizing the process in a proper system.
5. To speed up hiring decisions using automated shortlisting and interview scheduling.
6. To provide real-time data and reports to help companies make better decisions.

Initial Functional Requirements

FR01 - The system allows recruiters to create job posts, which will automatically update the job board.

FR02 - When a candidate submits an application, the system sends a confirmation message automatically.

FR03 - When the recruiter updates candidate status, the system will notify the candidate immediately.

FR04 - The system can filter resumes automatically based on criteria like skills and experience.

FR05 - Recruiters can manage candidates in one place, such as marking them as shortlisted or interviewed.

FR06 - The system helps schedule interviews and sends invitations to candidates.

FR07 - A dashboard is provided to show real-time recruitment progress and statistics.

Team Member Roles

Name	Role	Responsibilities
Nurul Iffah Sabrina Binti Ahmad Shukri	System Architect & Designer	Design system structure, create UML diagrams, and decide architecture
Izz Dhia Damia Binti Mat Zaidi	Backend & Integration Lead	Develop backend system, manage database, and handle API integration